Classical Field Theory — Exercise Sheet 25

Analytic extension and direct scattering

Solutions

Use $L = -\partial_x^2 + u$, the Wronskian $W(f,g) = fg' - f'g$, and $f_- = af_+(\cdot,-k) + bf_+(\cdot,k)$. For real $k \neq 0$, the right reflection amplitude is $r = b/a$. All potentials in this sheet are real Schwartz functions.

Exercise 25.1 (An upper-half-plane Volterra construction). For $\operatorname{Im} k \geq 0$ define

$$H(s,k) = \int_0^s e^{2ik\sigma} d\sigma, \qquad s \geq 0.$$

Thus $H(s,k) = (e^{2iks} - 1)/(2ik)$ for $k \neq 0$ and $H(s,0) = s$. Consider

$$m(x,k) = 1 + \int_x^\infty H(y-x,k)u(y)m(y,k)dy.$$

(a) For every finite R, construct the unique bounded solution on $x \geq R$ by iteration, uniformly for k in the closed upper half-plane. Prove that $m(x,k)$ is holomorphic in $\operatorname{Im} k > 0$ for each fixed x, and continuous up to its real boundary, including $k = 0$.

(b) Prove $m \to 1$ and $m_x \to 0$ as $x \to +\infty$. Show that $f_+(x,k) = e^{ikx}m(x,k)$ solves $Lf_+ = k^2 f_+$. State the two right asymptotic conditions at zero energy. Does this prove a nonzero Wronskian basis at $k = 0$?

The complex-analysis inputs may be used as givens: dominated parameter integrals of holomorphic functions are holomorphic, and locally uniform limits of holomorphic functions are holomorphic. These are the parameter-integral theorem and the uniform-limit result proved in *Analysis*, Lecture 10.

Solution. For $s \geq 0$, the defining integral gives $|H(s,k)| \leq s$ throughout the closed upper half-plane. Put $m_0 = 1$ and let m_j be the jth Volterra iterate. For $x \geq R$, each kernel factor at an ordered integration point y is at most $y - R$. Hence ordered integration yields

$$|m_j(x,k)| \leq \frac{W_R^j}{j!}, \qquad W_R = \int_R^\infty (y-R)|u(y)|dy < \infty.$$

The series $\sum_{j\geq 0} m_j$ converges uniformly in $x \geq R$ and in all k in that closed half-plane. Dominated convergence gives the integral equation. If a bounded difference satisfies its homogeneous equation, j iterations bound it by its supremum times $W_R^j/j!$. Letting $j \to \infty$ proves uniqueness.

For each finite list of ordered points the product of the kernels is holomorphic in k. Its integrable majorant is the product of the weights $(y-R)|u(y)|$, independent of k on the closed upper half-plane. The given parameter-integral theorem therefore makes every iterate holomorphic in the open upper half-plane. Their locally uniform sum is holomorphic. The same majorant and dominated convergence give continuity of every iterate on the closure, including $k = 0$. Uniform convergence of the series preserves that continuity. Solutions constructed with different R agree on overlaps by uniqueness, so they define m at every finite x.

The refined bound at a chosen x is

$$|m(x,k)-1| \le e^{W(x)}-1, \qquad W(x)=\int_x^\infty (y-x)|u(y)|\mathrm{d}y.$$

It tends to zero as $x \to +\infty$. Differentiating the integral with its lower endpoint gives

$$m_x = -\int_x^\infty e^{2\mathrm{i}k(y-x)}u(y)m(y,k)\mathrm{d}y, \qquad m_{xx} = um - 2\mathrm{i}km_x.$$

The first formula is justified by an integrable majorant on each right half-line; its modulus is at most $e^{W_R}\int_x^\infty |u(y)|\mathrm{d}y$, which tends to zero. The second follows from FTC and one more differentiation. Thus

$$f_+'' = e^{\mathrm{i}kx}(-k^2m + 2\mathrm{i}km_x + m_{xx}) = (u-k^2)f_+.$$

It has the relative tails $e^{-\mathrm{i}kx}f_+ \to 1$ and $e^{-\mathrm{i}kx}f_+' \to \mathrm{i}k$. At zero these say $f_+(x,0) \to 1$ and $f_+'(x,0) \to 0$. The construction includes this boundary value, but no pair with Wronskian $2\mathrm{i}k \neq 0$ exists by that formula at $k=0$. It proves neither a zero-energy scattering basis nor absence of a resonance.

Exercise 25.2 (Reflectionless scattering by intertwining). Fix $\kappa > 0$ and set

$$L_\kappa = -\partial_x^2 - 2\kappa^2\operatorname{sech}^2(\kappa x), \qquad A = \partial_x + \kappa\tanh(\kappa x), \quad A^\dagger = -\partial_x + \kappa\tanh(\kappa x).$$

(a) Factor $L_\kappa + \kappa^2$ and the free operator plus κ^2. Use the resulting intertwining identity to construct both real-energy Jost solutions and determine a, b, r with the fixed Wronskian convention.

(b) Find the right-normalised negative-energy state and its inverse squared norm. Show that there are no other negative-energy L^2 states. Then give a bounded zero-energy solution which is not in L^2.

Use the Sobolev integration-by-parts and domain arguments of Lecture 18. A bounded non-L^2 zero-energy solution is called a *zero-energy resonance* in this problem.

Solution. The product rule gives

$$A^\dagger A = L_\kappa + \kappa^2, \qquad AA^\dagger = L_0 + \kappa^2, \qquad L_0 = -\partial_x^2.$$

Consequently $L_\kappa A^\dagger = A^\dagger L_0$ as differential expressions on smooth functions. In this calculation a free exponential is not asserted to belong to the L^2 operator domain. Applying this to $e^{\mathrm{i}kx}$ and dividing by $\kappa - \mathrm{i}k$ gives

$$f_+(x,k) = e^{\mathrm{i}kx}\frac{k+\mathrm{i}\kappa\tanh(\kappa x)}{k+\mathrm{i}\kappa}.$$

Its right coefficient and derivative have the required limits. Reflection in x constructs the left solution

$$f_-(x,k) = e^{-\mathrm{i}kx}\frac{k-\mathrm{i}\kappa\tanh(\kappa x)}{k+\mathrm{i}\kappa}.$$

Since $f_+(x,-k) = e^{-\mathrm{i}kx}(k-\mathrm{i}\kappa\tanh(\kappa x))/(k-\mathrm{i}\kappa)$, comparison gives

$$f_- = \frac{k-\mathrm{i}\kappa}{k+\mathrm{i}\kappa}f_+(x,-k).$$

Thus $a = (k-\mathrm{i}\kappa)/(k+\mathrm{i}\kappa)$ and $b = r = 0$. This also gives $W(f_-,f_+) = 2\mathrm{i}ka$ with the stated order. For real k, $|a| = 1$, in agreement with the flux identity.

At $k = \mathrm{i}\kappa$ the right solution is

$$f_+(x, \mathrm{i}\kappa) = \frac{1}{2} e^{-\kappa x}(1 + \tanh(\kappa x)) = \frac{1}{2}\operatorname{sech}(\kappa x).$$

It decays at both ends, has right coefficient one and squared norm $\frac{1}{4}\int \operatorname{sech}^2(\kappa x)\mathrm{d}x = 1/(2\kappa)$. Hence its inverse squared norm is 2κ. It is in the kernel of A, so its energy is $-\kappa^2$.

If $L_\kappa g = -\mu^2 g$ with $\mu > 0$ and $g \in H^2$, its equation and the bounded smooth potential bootstrap g to every H^m. The intertwining $AL_\kappa = L_0 A$ gives $L_0(Ag) = -\mu^2 Ag$ with $Ag \in H^2$. Free integration by parts implies $\|(Ag)'\|_2^2 = -\mu^2\|Ag\|_2^2$, so $Ag = 0$. Solving this first-order equation gives $g = C\operatorname{sech}(\kappa x)$, and its energy is necessarily $-\kappa^2$. This proves uniqueness of the negative level and its one-dimensional eigenspace.

Finally, letting $k \to 0$ in the right solution gives $\tanh(\kappa x)$. Direct differentiation also verifies $L_\kappa \tanh(\kappa x) = 0$. Its two endpoint limits are -1 and 1, so it is bounded but not in L^2. It is a resonance under the stated definition, not an additional bound state.

Exercise 25.3 (A weak asymmetric potential retains a phase). Let $u_\epsilon = \epsilon V$, with $\epsilon \in \mathbb{R}$, and

$$V(x) = e^{-(x-1)^2} + 2e^{-(x+2)^2}.$$

For each fixed real $k \neq 0$, derive the first-order expansions of a, b, r as $\epsilon \to 0$ and prove an $O(\epsilon^2)$ remainder bound. Evaluate the Gaussian coefficient and the leading reflection probability. Show that no negative-energy bound states occur when $\epsilon \geq 0$. The expansion is a fixed-k assertion; no uniform limit at $k = 0$ is requested.

Solution. The left Volterra equation is

$$f_-(x, k) = e^{-\mathrm{i}kx} + \int_{-\infty}^{x} \frac{\sin k(x-y)}{k}\, \epsilon V(y) f_-(y, k)\mathrm{d}y.$$

The same ordered iteration as Lecture 25 gives, on the whole line,

$$|f_-| \leq e^{\delta}, \qquad |f_- - e^{-\mathrm{i}kx}| \leq e^{\delta} - 1, \qquad \delta = \frac{|\epsilon|\|V\|_1}{|k|}.$$

Separating the sine and taking the two endpoint coefficients at the right gives the exact integral expressions

$$a = 1 - \frac{\epsilon}{2\mathrm{i}k}\int_{\mathbb{R}} e^{\mathrm{i}ky} V(y) f_-(y, k)\mathrm{d}y, \qquad b = \frac{\epsilon}{2\mathrm{i}k}\int_{\mathbb{R}} e^{-\mathrm{i}ky} V(y) f_-(y, k)\mathrm{d}y.$$

The derivative asymptotics follow from the differentiated Volterra equation, so these are the coefficients in the actual right Jost basis. Replacing f_- by its free term makes an error in each coefficient at most

$$\frac{|\epsilon|\|V\|_1}{2|k|}(e^{\delta} - 1) = O(\epsilon^2).$$

It follows that

$$a = 1 - \frac{\epsilon}{2\mathrm{i}k}\int V + O(\epsilon^2), \qquad b = \frac{\epsilon}{2\mathrm{i}k}\int e^{-2\mathrm{i}ky} V(y)\mathrm{d}y + O(\epsilon^2).$$

Since $a = 1 + O(\epsilon)$ and $a \neq 0$, division gives the same first-order coefficient for $r = b/a$.

Translation of the Gaussian integral yields

$$\int V = 3\sqrt{\pi}, \qquad \int e^{-2\mathrm{i}kx} V(x)\mathrm{d}x = \sqrt{\pi} e^{-k^2}(e^{-2\mathrm{i}k} + 2e^{4\mathrm{i}k}).$$

Thus

$$a = 1 - \frac{3\epsilon\sqrt{\pi}}{2\mathrm{i}k} + O(\epsilon^2),$$

$$r = \frac{\epsilon\sqrt{\pi}e^{-k^2}}{2\mathrm{i}k}(e^{-2\mathrm{i}k} + 2e^{4\mathrm{i}k}) + O(\epsilon^2).$$

and b has the same first-order term. The absolute square of the parenthesis is $5 + 4\cos 6k$. Squaring the amplitude expansion gives

$$|r|^2 = \frac{\epsilon^2\pi e^{-2k^2}}{4k^2}(5 + 4\cos 6k) + O(|\epsilon|^3).$$

For instance at $k = \pi/4$ the parenthesis is $-2 - \mathrm{i}$, so its coefficient after division by i has both real and imaginary parts. Taking the probability discards that phase.

For $\epsilon \geq 0$ the potential is nonnegative. A putative bound state $Lf = -\mu^2 f$ would give

$$\int (|f'|^2 + \epsilon V|f|^2)\mathrm{d}x = -\mu^2\|f\|_2^2,$$

which is impossible for $\mu > 0$ and nonzero f. This proves the stated absence without asserting that weak negative potentials have the same property.